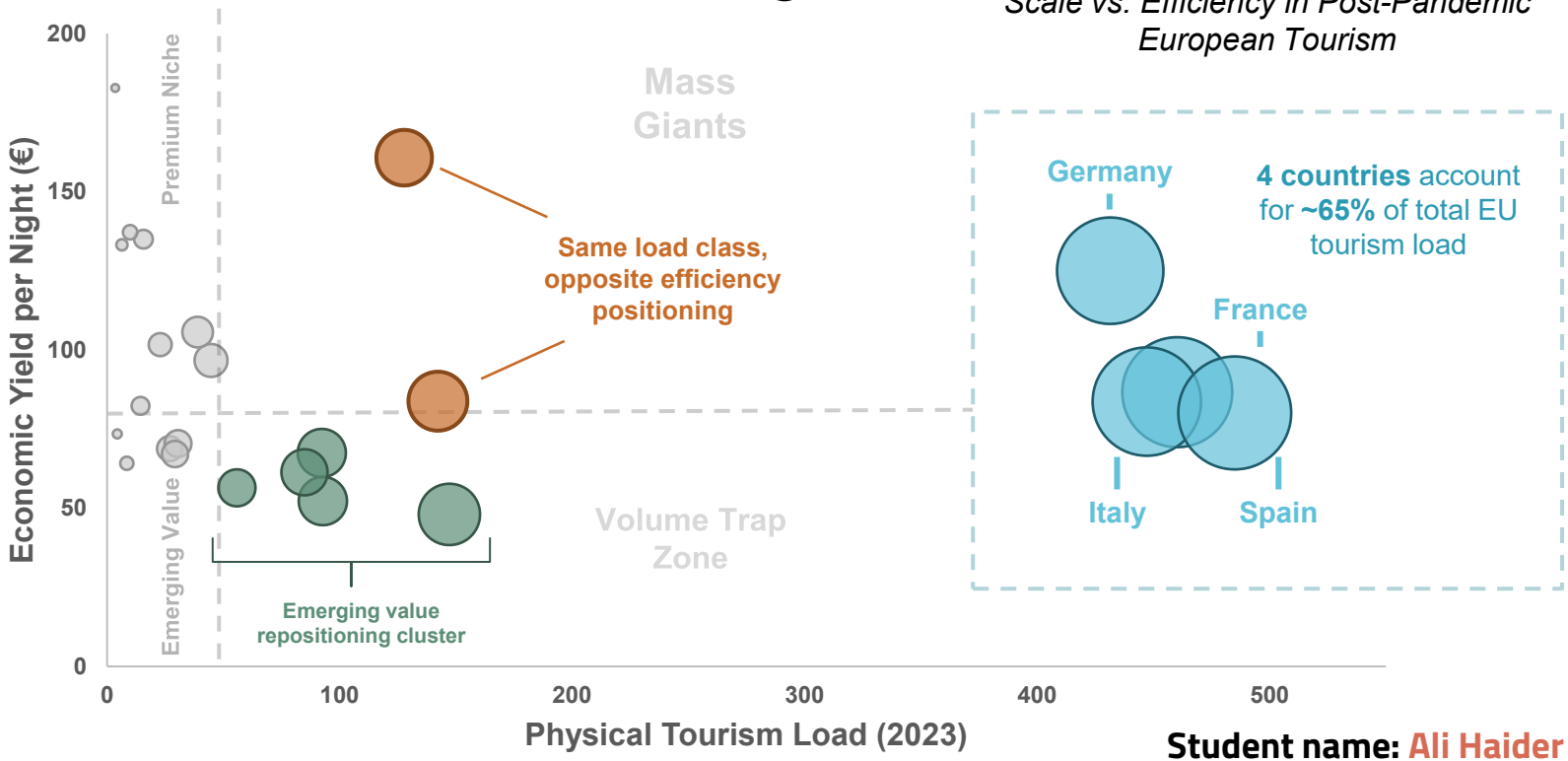
Data Visualization Pitch

Student Individual Assignment

THE VOLUME TRAP

Scale vs. Efficiency in Post-Pandemic European Tourism



*Quadrants defined using 2023 median EU tourism load (42M nights) and median yield (€83)
#Expenditure values are nominal and not inflation-adjusted
Source: Eurostat – Nights spent (tour_occ_ninat) and Average expenditure per night (tin00196), 2019–2023

Student name: Ali Haider
Student ID: 948092
Student e-mail address: a.haider16@campus.unimib.it

Is Europe optimizing for volume or efficiency in its post-pandemic tourism recovery?

Operational Definitions

Scale → Total nights spent (2023)

Efficiency → Average expenditure per night (2023)

Structural Position → Assessed relative to EU median thresholds (2023)

Diagnostic Sub-Questions

- Do high-load destinations generate proportionally higher economic yield?
- Which countries improved or deteriorated in yield relative to 2019?
- How concentrated is Europe's tourism load across member states?

Project Overview

- Evaluate whether Europe's post-pandemic tourism recovery is structurally aligned with economic efficiency
- Compare total nights spent with average expenditure per night across EU member states
- Assess whether high tourism volume translates into proportional value capture
- Identify whether scale concentration masks underlying structural inefficiencies

The objective is to evaluate whether scale translates into value or whether structural inefficiencies persist

A clean, comparable dataset underpins the volume-yield diagnostic

Data Sources & Scope

Eurostat Tourism Database

- **tour_occ_ninat**: Nights spent at tourist accommodation establishments
- **tin00196:** Average expenditure per night

Geographic & Analytical Scope

- **Coverage:** EU member states
- **Aggregation level:** National (NUTS-0)
- Only countries with complete 2019–2023 observations were retained

Temporal Coverage & Units

- **Years analyzed:** 2019 (pre-pandemic baseline) and 2023 (post-pandemic recovery)
- Nights spent reported in absolute values (converted to millions for visualization)
- Expenditure per night measured in € (nominal values)

Data Engineering & Derived Metrics

Data Cleaning

- Removed incomplete country-year observations
- Ensured consistent country coverage across both indicators
- Standardized numeric formats and removed Eurostat flags

Feature Engineering

Post-COVID Efficiency Delta → % change in average expenditure per night (2019–2023)

2023 European Load Share (%) → Country nights ÷ Total EU nights (2023)

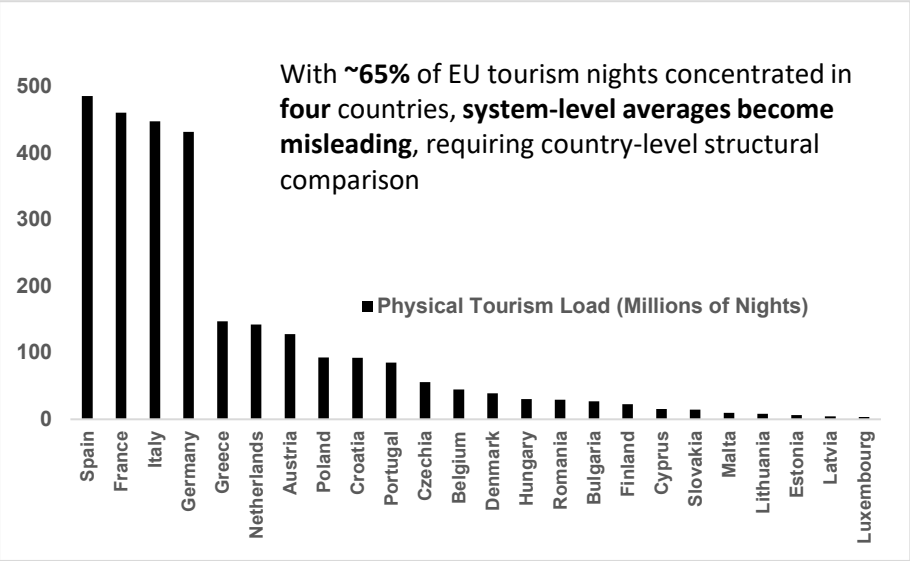
EU Median Thresholds (2023)→ Load: 42M nights | Yield: €83 per night

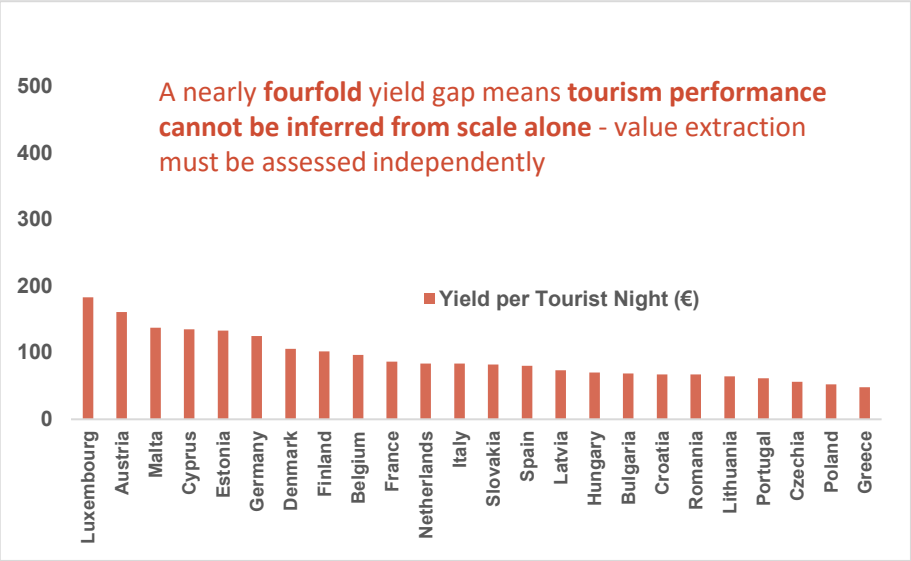
**Medians used to define structural quadrants in the Efficiency Matrix*

Integration Logic

- Volume and yield datasets merged using country–year keys
- National aggregation selected to ensure 1:1 comparability across indicators

When scale is concentrated and yield is dispersed, simple rankings no longer suffice





Structural Observations (2023)

- Tourism load is highly concentrated among a small number of member states
- Economic yield per night exhibits substantial cross-country dispersion
- Scale and efficiency are unevenly distributed across the EU tourism system

When volume is concentrated and yield is dispersed, structural positioning - not absolute magnitude - becomes the relevant analytical lens

Post-COVID recovery redistributed efficiency leadership across Europe - exposing structural flexibility in emerging markets and saturation constraints in mature ones.

Top 3 countries

Country	% Post COVID Efficiency Delta
Denmark	-26%
Netherlands	-4%
Malta	-3%

Bottom 3 countries

Country	% Post COVID Efficiency Delta
Hungary	75%
Bulgaria	62%
Romania	55%

Key Insights (Condensed)

- **Efficiency gains were concentrated in Central & Eastern Europe**, led by Hungary, Bulgaria and Romania
- **Mature Western markets showed limited structural improvement**, with Denmark and the Netherlands experiencing efficiency erosion
- **Scale did not determine performance** - large tourism systems did not translate recovery into proportional efficiency gains

Volume recovery ≠ value capture

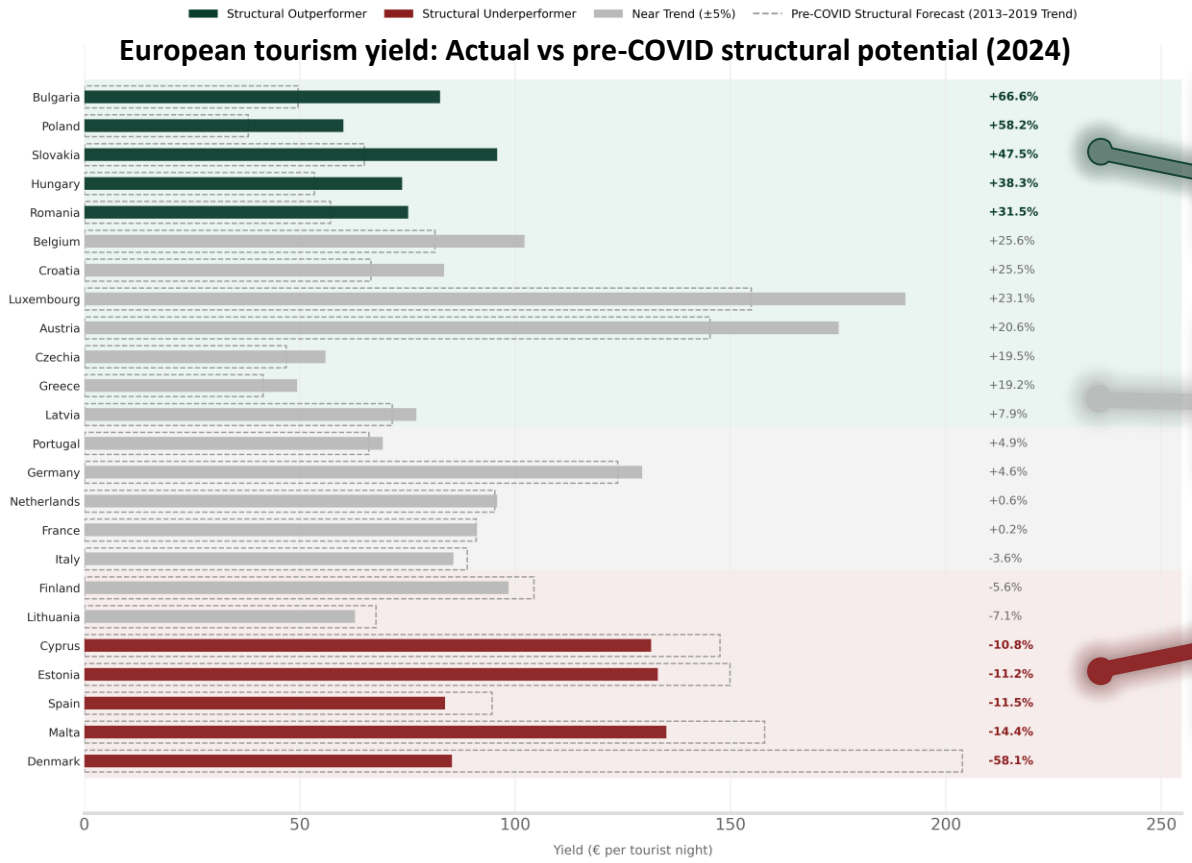
Post-COVID success depended on structural repositioning, not traffic alone

Regional efficiency shift (mean vs. volume-weighted)	
Central & Eastern	46% (weighted **46%**)
Southern / Mediterranean	18% (weighted **17%**)
Western Core	9% (weighted **12%**)
Nordics	-11% (weighted **-15%**)

Post COVID Efficiency Delta

-26% 25% 75%

The post-pandemic landscape has redistributed efficiency leadership: rewarding structural flexibility while exposing saturation in mature, high-volume markets



Diagnostic note: Per-country linear regression trained on 2013–2019 Eurostat yield baseline (n=7). 2024 actuals compared against counterfactual trend projection.

Eastern Structural Reset
Central & Eastern Europe has fundamentally broken from its historical convergence path, undergoing a permanent pricing regime change.

Mature Market Equilibrium
Western Core economies are tracking almost perfectly along their 2013–2019 structural trajectories. Mature markets preserve value but struggle to expand it

Scale Penalty Zone
Despite restored traffic levels, high-volume destinations remain materially below their modeled structural potential. Scale recovery is no longer a guarantee of economic extraction.

Europe’s recovery has revealed a structural truth: Volume restores activity, but only flexibility restores value

Efficiency Leadership Is Migrating East

CEE markets demonstrate unprecedented structural pricing flexibility, outperforming pre-COVID trajectories by up to 65%.

- Investment and policy should prioritize premium repositioning over volume expansion. These are now the "Efficiency Blueprints" for the continent.

The Maturity Cap in the Western Core

Western and Nordic systems appear structurally capped, tracking almost exactly to 2013–2019 trends.

- Marginal volume gains in these saturated markets are likely to erode pricing power. The focus must shift to "Yield Protection" rather than "Growth."

The End of the Scale Advantage

Massive tourism systems (The Mediterranean Giants) show the most significant "Efficiency Rupture" (Gap between potential and actual yield).

- Future competitiveness depends on yield optimization (extracting more from fewer) to escape the "Volume Trap."

Statistical rigor is maintained through a counterfactual framework, ensuring that structural insights remain valid despite external macroeconomic volatility

Data Treatment & Sample Integrity

- **The "Pairwise Deletion" Rule:** 13 countries were excluded from the final diagnostic to ensure 100% longitudinal consistency between the 2013-2019 baseline and 2024 actuals.
- **Source:** All data is harmonized from **Eurostat (NACE Rev. 2)** to ensure a single "Source of Truth" for physical load and economic yield.

Statistical Assumptions

- **The Linearity Constraint:** The 2024 "Ghost" forecast utilizes a **Linear Regression** model. This assumes that tourism yields would have continued a stable, linear trajectory if the COVID-19 shock had not occurred.
- **Counterfactual Nature:** This is a counterfactual analysis - it measures the **gap** created by a systemic rupture, rather than predicting future market fluctuations.

Macroeconomic Variables

- **Nominal vs. Real Values:** Yield calculations are based on **nominal Eurostat figures**. They do not explicitly adjust for the 2022-2023 inflationary spikes across the Eurozone, which may slightly inflate the "Efficiency Miracle" in high-inflation regions.
- **Currency Fluctuations:** For non-Eurozone countries (e.g. Hungary, Poland), yield is subject to exchange rate volatility relative to the Euro.

Granularity & Scope

- **National Aggregation:** Using country-level data may mask high-performing sub-national clusters (e.g., the "Milan Effect" vs. the "Italy Average").
- **Elasticity:** The current model does not control for demand elasticity or specific accommodation shifts (e.g., the rise of P2P rentals vs. traditional hotels).

LICENCE

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Ali Haider

Data Scientist | Industrial & Applied Analytics

MSc Data Science

Università degli Studi di Milano-Bicocca

Core Expertise

- Predictive modelling and statistical analysis using regression and clustering techniques
- Large-scale data analysis using SQL and Python across structured datasets
- Feature engineering and development of custom analytical indicators
- Data visualization and analytical storytelling using Python (Matplotlib, Seaborn)
- Exploratory data analysis and pattern discovery in complex datasets
- Building analytical frameworks to diagnose structural patterns and performance drivers

Professional Focus

- Applying data science to large-scale operational and economic systems
- Industrial and energy systems analytics
- Infrastructure efficiency and resource utilization diagnostics
- Decision-support modeling and optimization-driven analytics

basicalihaider

a.haider16@campus.unimib.it *(college)*
basicalihaider@gmail.com *(personal)*

LinkedIn